Benchmarking Forecast-Ranking Robustness Under Frictional Deployment

Abstract

Forecasting benchmarks often rank systems by forecast-side quality. Under frictional deployment, however, forecasts pass through a fixed interface before actions and utility are realized, so the ranking appropriate for prediction need not be the ranking appropriate for deployed model selection. We therefore study Q2 as the main benchmark question—whether forecast-side rankings remain deployed-selection robust under a fixed frictional interface—while using Q1 only as mechanism support for why divergence can arise. Synthetic provides the exact zero-friction sanity anchor; event-micro provides the main forecasting-native evidence, while inventory provides the main operational corroboration, and recurrent deployed misselection appears at moderate-to-high friction. Forecast-side metrics may remain valid for prediction, but forecasting systems under frictional deployment interfaces should report deployed-selection robustness, not forecast-side ranking alone.

1 Introduction

Forecasting benchmarks often evaluate systems by forecast-side quality alone. Under deployment, however, forecasts are translated through fixed interfaces into realized actions and utility. The benchmark-design question is therefore not only whether predictions are accurate, but whether forecast-side ranking remains selection-robust once deployment friction intervenes. A benchmark is unreliable for deployment if the forecast-side winner recurrently fails to be the deployed-best system under the fixed interface that is actually used.

We study this issue through two evaluation questions. Q2 is the main benchmark question: under one fixed deployed interface, does ranking systems by forecast quality still identify the system that is best under realized-action evaluation? Q1 is mechanism support only: if a proposed action path is held fixed, can different interfaces produce different realized outcomes? This separation keeps forecast metrics in their proper role as prediction measures while isolating when the deployment-time selection object changes under friction.

Forecasting benchmark work often evaluates forecast quality, calibration, and live predictive performance. Our question is complementary: whether forecast-side ranking remains trustworthy for deployed model selection under a fixed frictional interface. Unlike decision-aware forecasting or control work, we do not propose new training objectives or policies; we study the evaluation object used for model selection after forecasts are mediated by actions and friction.

This paper makes three narrow contributions. First, it frames deployed-selection robustness as a benchmark-design object for forecasting systems under fixed frictional interfaces. Second, it documents the Q2 failure mode: forecast-side winners can become recurrently deployed-suboptimal at moderate-to-high friction under one fixed interface. Third, it uses Q1 only to explain why forecast-side, target-side, and realized-action evaluation can diverge while keeping forecast-side metrics in their role as valid prediction measures. Evidence comes from a synthetic exact zero-friction sanity anchor, a minimal forecasting-native event-micro benchmark evaluated over 100 seeds, inventory as the main operational corroboration, and load-following retained only as appendix-only secondary corroboration.

2 Two Evaluation Questions

Let $\hat{y}_t$ denote a forecast, a_t^{tar} a proposed action derived from it, and a_t^{exec} the realized action after a frictional interface is applied. Let $U(a, y)$ denote realized operational utility against the realized outcome y_t. We distinguish three evaluation objects: forecast-side quality, target-side decision quality, and realized-action quality. The benchmark question is which of these objects is appropriate for model selection under deployment constraints.

Q1 (same proposal, different interface). Holding the proposal fixed at $a_{1:T}^{tar}$, we ask whether varying only the interface that maps proposals to realized actions changes realized operational outcomes. This is an exact-control question: proposal identity must be locked, so any downstream difference must arise from the interface rather than from a different forecast or proposal source.

Q2 (fixed interface, different forecasts). Q2 is the benchmark-selection question. Holding the interface fixed, we ask whether varying the forecaster changes the relationship between forecast-metric ranking and deployed operational ranking. More directly, does ranking systems by forecast quality still identify the system that is best under deployed realized-action evaluation?

These questions are complementary but not interchangeable. Q1 explains why divergence can arise once realized actions depart from proposed targets, while Q2 shows when that divergence becomes a benchmark model-selection failure under a fixed deployed interface. Because the two questions change different parts of the pipeline, evidence for one should not be read as automatic evidence for the other. A short appendix proposition formalizes why forecast-side ordering need not be preserved after frictional execution, even when forecast-side metrics may remain valid for prediction.

3 Evidence Design

Our evidence hierarchy is Q2-first. Synthetic provides the exact zero-friction sanity anchor. A minimal event-micro benchmark evaluated over 100 seeds provides the main forecasting-native Q2 evidence: models emit event probabilities, a fixed thresholding rule maps probabilities to actions, and switching

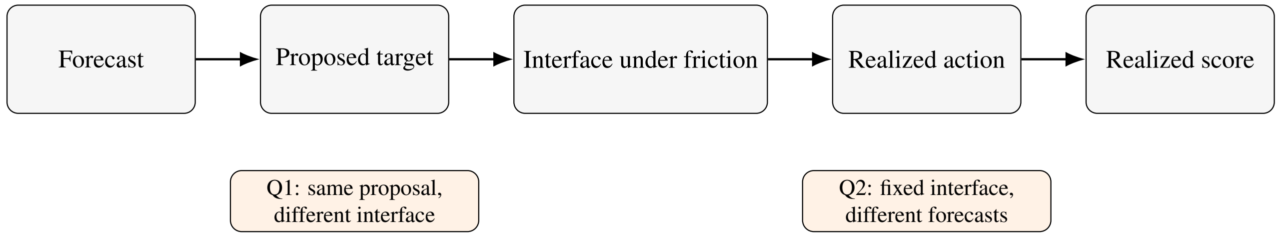

Figure 1: Forecasts are converted into proposed target actions and then translated into realized actions through an interface operating under friction, which creates two distinct evaluation questions: whether the same frozen proposal can yield different realized outcomes under different interfaces (Q1), and whether forecast-metric ranking matches deployed operational ranking under a fixed interface (Q2). This separation changes the evaluation object and therefore can change both benchmark interpretation and deployed model selection under friction.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive sharp	Reactive sharp	0.62	0.003	38/100
0.50	Reactive sharp	Calibrated baseline	0.31	0.011	69/100
1.00	Reactive sharp	Lagged smoother	0.01	0.057	99/100

Table 1: Forecast-side vs. deployed-side model selection in the forecasting-native event-micro benchmark. Under a fixed thresholding interface with switching friction, the forecast-selected model becomes recurrently deployed-suboptimal as friction rises.

friction penalizes action changes. Inventory provides the main operational corroboration, while load-following is retained only as appendix-only secondary corroboration because its zero row remains mixed. Q1 appears only as mechanism support from synthetic and inventory and is summarized briefly in the main text.

In inventory Q2, the locked main table compares five forecasting families under one responsive replenishment rule: Naive persistence, Moving average (7), Linear AR, Small MLP, and Small GRU. The appendix widens that comparison with validation-selected family representatives under the same split, held-out protocol, validation criterion, and standardized two-stage tuning procedure; simple heuristic baselines remain fixed ex ante. The main text keeps the locked five-family comparison, while the appendix records the broader family sweep as a stricter-gate transparency layer rather than as additional headline evidence.

4 Results

4.1 Q2: Fixed interface, benchmark failure and selection consequence

Q2 asks the paper's headline benchmark-design question: under one fixed deployed interface, can forecast-side ranking still be trusted for deployed model selection? Synthetic provides the exact zero-friction sanity anchor: rankings align exactly at zero friction, and once friction is moderate or high, repeated pairwise flips and lower rank correlation emerge.

The event-micro table provides the main forecasting-native evidence. In a minimal event-micro benchmark evaluated over 100 seeds, agreement is highest at zero friction (0.62),

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Small MLP	0.70	0.022	3/10
0.25	Small MLP	Linear AR	0.70	0.053	3/10
0.50	Small MLP	Moving average (7)	0.10	0.308	9/10
1.00	Small MLP	Moving average (7)	0.00	1.187	10/10

Table 2: Operational corroboration under a fixed replenishment interface.

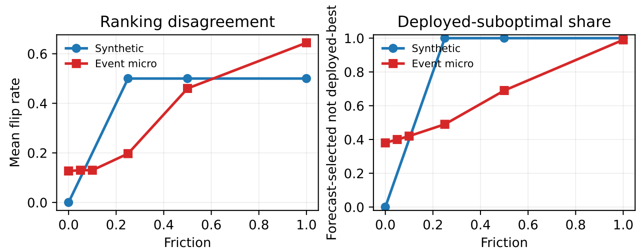

Figure 2: Q2 — synthetic provides the exact zero-friction sanity anchor, while event micro provides the paper's main forecasting-native evidence under a fixed thresholding interface. The left panel shows pairwise ranking disagreement (mean flip rate); the right panel shows the operational benchmark-consequence quantity, the share of seeds in which the forecast-selected model is not deployed-best. Appendix uncertainty bands and seed-level recurrence checks are reported separately.

falls to 0.31 by 0.50, and reaches 0.01 at 1.00. Over the same rows, the deployed winner shifts from Reactive sharp to Calibrated baseline and then Lagged smoother, while deployed-suboptimality reaches $69/100$ seeds at 0.50 and $99/100$ at 1.00. The event-micro pattern persists under an alternate threshold, under a hysteresis-threshold interface, and when forecast-side ranking is recomputed with log loss.

The inventory table provides operational corroboration with stronger recurrence magnitudes. The friction 0.25 row is best read as a transition regime: the most frequent deployed winner changes to Linear AR, yet agreement remains 0.70, deployed suboptimality remains $3/10$, and the median deployed gap remains 0.0. By friction 0.5, however, benchmark failure is already recurrent: the forecast-selected model is deployed-suboptimal in $9/10$ seeds, agreement falls to 0.10, and the median deployed gap reaches 0.227. At friction 1.0, deployed suboptimality rises to $10/10$ seeds, with mean and median deployed gaps of 1.187 and 1.162. Thus the practical issue is not merely that rankings drift, but that model choice under a fixed deployed rule becomes repeatedly wrong at moderate-to-high friction.

An appendix-only load-following check remains directionally consistent but retains a mixed zero row, so we do not use it as headline corroboration in the main text (Appendix E).

Appendix Tables A3–A4 and Figures A2–A3 support the interpretation that the main moderate/high-friction patterns, especially in event-micro and inventory, are not driven solely by isolated seed noise.

Appendix Tables A12–A14 widen the inventory comparison set under a stricter standardized package gate. That broader-family sweep strengthens moderate/high-friction mismatch, but the zero- and low-friction rows remain too mixed for headline use and therefore do not overturn the narrower main claim from the locked five-family table.

Q1 mechanism support. Q1 is mechanism support rather than the paper's claim. Holding the proposed action path fixed, synthetic and inventory show that frictional interfaces can separate proposed and realized actions: synthetic yields an increasing proposal-versus-realization gap away from zero friction, while inventory shows a threshold pattern in which sufficiently high friction favors tempered execution. This explains why forecast-side, target-side, and realized-action evaluation can diverge; Q2 shows when that divergence becomes deployed misselection.

5 Discussion and Limitations

The paper's practical claim is narrow: forecast-side metrics may remain valid for prediction, yet under a fixed deployed interface they need not be sufficient for deployed model selection. Event-micro provides the main forecasting-native Q2 evidence, inventory provides the main operational corroboration, and Q1 explains why the evaluation object can change under friction.

Three limitations bound the claim. Forecasting-native Q2 evidence comes from a minimal event-micro benchmark evaluated over 100 seeds, so the paper still relies on one narrow forecasting-native task family rather than a broad benchmark suite. Inventory anchors the paper's operational corroboration, while load-following remains appendix-only secondary corroboration because its zero row stays mixed even after grouping repair. In the expanded-family inventory sweep, moderate/high-friction mismatch strengthens but zero- and low-friction rows remain mixed, so we do not use that stricter-gate transparency block as main-text confirmation.

The resulting implication is evaluative rather than predictive: forecasting systems under deployment constraints should be judged with realized-action-aware criteria rather than forecast-side quantities alone.

A Evidence Hierarchy

Domain	Question	Fixed	Varying	Zero-fric.	Positive-fric.	Role
Synthetic	Q2	interface	forecaster	aligned at zero	ranking flips	zero-fric. anchor
Event micro	Q2	interface	forecaster	mixed at zero	selection failure recurs	main fcst.-native Q2 evid.
Inventory	Q2	interface	forecaster	mixed at zero	stronger recurrence	main oper. corrob.
Load-following	Q2	interface	forecaster	mixed at zero	directional recurrence	appx.-only sec. corrob.
Synthetic	Q1	proposal	interface	exact agreement	mismatch emerges	mech. support
Inventory	Q1	proposal	interface	coincide at zero	threshold story	mech. support

Table A1: Evidence map for the paper's Q2-first hierarchy, covering the synthetic, event-micro, inventory, and load-following evidence blocks used in the paper and appendix.

B Q1 Mechanism Support

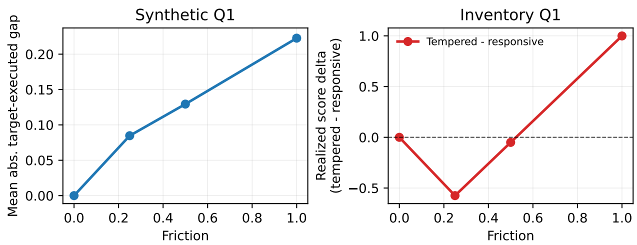

Figure A1: Appendix Q1 mechanism support. Synthetic yields exact zero-friction coincidence and a rising positive-friction proposal-versus-realization gap, while inventory shows a threshold pattern in which sufficiently high friction favors tempered execution. The figure explains why Q2 divergence can arise but is not itself the paper's headline evidence.

C Additional Q2 Package Details

This appendix collects the cross-domain seed-level reliability checks and a stricter same-interface package-gate transparency layer after the fifth inventory baseline is added. These appendix-only additions do not overturn the narrower main claim from the locked five-family inventory table.

The same locked Q2 package also admits simple seed-level reliability checks. Table A3 reports one-sided exact binomial tests for whether the forecast-selected model is deployed-suboptimal in a majority of seeds in the main recurrence rows. Table A4 reports 95% bootstrap intervals for the deployed-gap rows. Figure A2 adds uncertainty bands for the synthetic and event-micro Q2 curves, while Figure A3 shows seed-level disagreement strips for the main Q2 domains.

Domain	Friction	Deployed-suboptimal seeds / total	Share	95% exact CI	One-sided binomial p
Event micro	0.50	69/100	0.69	[0.605, 1.000]	<0.001
Event micro	1.00	99/100	0.99	[0.953, 1.000]	<0.001
Inventory	0.50	9/10	0.90	[0.606, 1.000]	0.011
Inventory	1.00	10/10	1.00	[0.741, 1.000]	<0.001
Load-following	0.25	7/10	0.70	[0.393, 1.000]	0.172
Load-following	0.50	7/10	0.70	[0.393, 1.000]	0.172
Load-following	1.00	8/10	0.80	[0.493, 1.000]	0.055

Table A3: One-sided exact binomial tests for whether the forecast-selected model is deployed-suboptimal in a majority of seeds. The strongest event-micro and inventory rows show clear majority recurrence, while load-following shows the same directional pattern with smaller seed counts.

Domain	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Event micro	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Event micro	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Inventory	0.50	0.308	[0.186, 0.443]	0.227	[0.176, 0.484]
Inventory	1.00	1.187	[0.995, 1.390]	1.162	[0.969, 1.389]
Load-following	0.25	0.001	[0.000, 0.001]	0.000	[0.000, 0.001]
Load-following	0.50	0.001	[0.000, 0.002]	0.001	[0.000, 0.002]
Load-following	1.00	0.002	[0.001, 0.005]	0.002	[0.000, 0.003]

Table A4: 95% bootstrap intervals for deployed-gap rows in the main Q2 evidence blocks. Event-micro and inventory provide the strongest positive-gap support, while load-following remains directional secondary corroboration.

Domain	Role	Status	Seeds	Zero flip	Zero rho	Best +fric.	Best +flip	Strongest pair	Flip share	Note
Synthetic	required	clears stricter gate	20	0.000	1.000	0.2500	0.500	Linear AR / Reactive heuristic	1.000	passed stricter gate
Inventory	required	mixed zero row	10	0.120	0.840	1.0000	0.500	Small MLP / Moving average (7)	1.000	does not clear stricter gate

Table A2: Stricter same-interface package-gate summary for the appendix transparency layer. These statuses do not overturn the narrower main-text claim from the locked five-family inventory table.

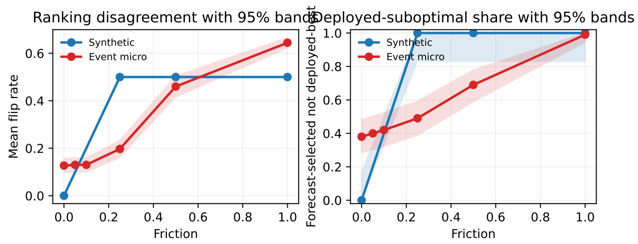

Figure A2: Appendix uncertainty bands for the synthetic and event-micro Q2 curves. Left: mean flip rate with 95% bands from seed-level standard errors. Right: deployed-suboptimal share with exact 95% binomial intervals.

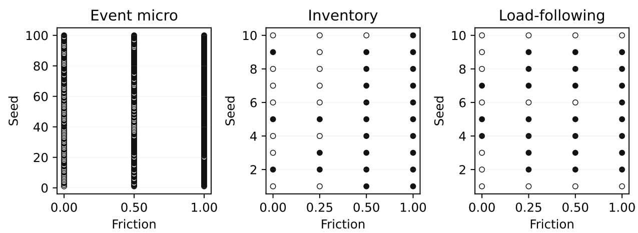

Figure A3: Seed-level disagreement strips for the main Q2 domains. Filled markers indicate seeds where the forecast-selected model is not deployed-best under the fixed interface.

D Event-Micro Robustness Package

Because event-micro provides the paper's main forecasting-native Q2 evidence, this appendix records the first domain-specific robustness package for that benchmark. We construct a minimal forecasting-native binary-event setting evaluated over 100 seeds, in which each forecaster emits event probabilities, a fixed thresholding rule maps probabilities to actions, and switching friction penalizes action changes. The main text uses Brier as the canonical forecast-side criterion. This appendix records the full six-row Brier table plus an alternate-threshold check, a log-loss reranking check, and a second-interface hysteresis check.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap
0.00	Reactive sharp	Reactive sharp	0.62	0.003
0.05	Reactive sharp	Reactive sharp	0.60	0.002
0.10	Reactive sharp	Reactive sharp	0.58	0.002
0.25	Reactive sharp	Calibrated baseline	0.51	0.003
0.50	Reactive sharp	Calibrated baseline	0.31	0.011
1.00	Reactive sharp	Lagged smoother	0.01	0.057

Table A5: Forecast-side vs. deployed-side model selection in the full six-row event-micro benchmark summary. Under the canonical fixed-threshold interface, deployed-suboptimal counts rise from $38/100$ seeds at zero friction to $99/100$ at friction 1.00.

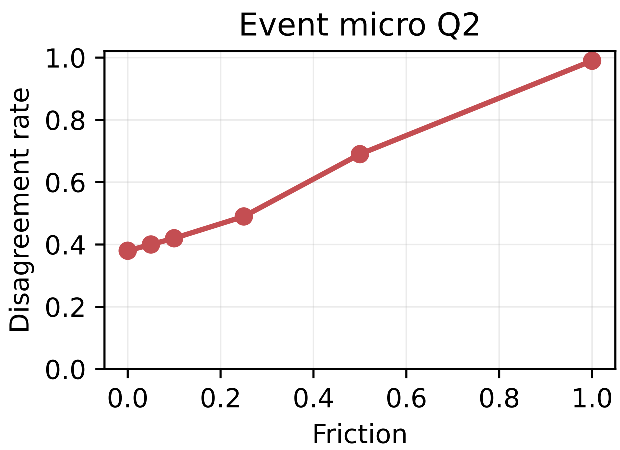

Figure A4: Ranking mismatch in a minimal event-forecasting micro-benchmark. Disagreement between forecast-side and deployed-side model selection is lowest at zero and very low friction and increases as switching frictions rise.

Threshold	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
tau=0.55	0.00	Reactive sharp	Reactive sharp	0.62	0.003	0.000	38/100
tau=0.55	0.50	Reactive sharp	Calibrated baseline	0.31	0.011	0.008	69/100
tau=0.55	1.00	Reactive sharp	Lagged smoother	0.01	0.057	0.053	99/100
tau=0.50	0.00	Reactive sharp	Reactive sharp	0.56	0.002	0.000	44/100
tau=0.50	0.50	Reactive sharp	Calibrated baseline	0.23	0.014	0.011	77/100
tau=0.50	1.00	Reactive sharp	Lagged smoother	0.02	0.061	0.059	98/100

Table A6: Alternate-threshold robustness for the event-micro benchmark. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when the fixed threshold is changed from $\tau = 0.55$ to $\tau = 0.50$.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive sharp	Reactive sharp	0.62	0.002	0.000	38/100
0.50	Reactive sharp	Calibrated baseline	0.26	0.013	0.010	74/100
1.00	Reactive sharp	Lagged smoother	0.01	0.060	0.057	99/100

Table A7: Log-loss robustness for the event-micro benchmark at the canonical threshold. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when forecast-side ranking is recomputed with log loss instead of Brier.

Interface	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
Hysteresis threshold	0.00	Reactive sharp	Reactive sharp	0.55	0.003	0.000	45/100
Hysteresis threshold	0.50	Reactive sharp	Calibrated baseline	0.45	0.005	0.001	55/100
Hysteresis threshold	1.00	Reactive sharp	Lagged smoother	0.18	0.022	0.020	82/100

Table A8: Second-interface robustness for the event-micro benchmark. Under a fixed hysteresis-threshold interface with $\delta = 0.05$, the same qualitative winner drift and deployed-suboptimality pattern remains visible, though the magnitudes are smaller than under the canonical fixed threshold.

Across this appendix robustness package, the same qualitative winner-drift and recurrent deployed-suboptimality pattern persists in a minimal forecasting-native event-micro benchmark evaluated over 100 seeds.

D.1 Interface-Aware Event-Micro Stats Addendum

Because the hysteresis-threshold check is retained paper-facing as a second-interface robustness result, this addendum records interface-specific recurrence and gap summaries without changing the main cross-domain recurrence package.

Interface	Friction	Deployed-suboptimal seeds / total	Share	95% exact CI	One-sided binomial p
Fixed threshold	0.50	69/100	0.69	[0.605, 1.000]	<0.001
Fixed threshold	1.00	99/100	0.99	[0.953, 1.000]	<0.001
Hysteresis threshold	0.50	55/100	0.55	[0.463, 1.000]	0.184
Hysteresis threshold	1.00	82/100	0.82	[0.745, 1.000]	<0.001

Table A9: Interface-aware recurrence tests for event-micro deployed-suboptimality. The canonical fixed threshold shows stronger recurrence, while hysteresis preserves the same direction with smaller magnitudes.

Interface	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Fixed threshold	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Fixed threshold	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Hysteresis threshold	0.50	0.005	[0.004, 0.006]	0.001	[0.000, 0.003]
Hysteresis threshold	1.00	0.022	[0.019, 0.026]	0.020	[0.015, 0.025]

Table A10: Interface-aware bootstrap intervals for the event-micro deployed gap. Positive intervals remain visible under both interfaces, with smaller gaps under hysteresis.

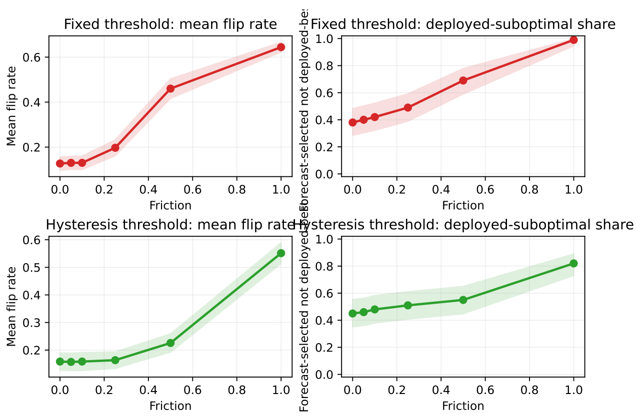

Figure A5: Interface-aware event-micro uncertainty addendum. Rows facet the canonical fixed threshold and the hysteresis threshold; columns show mean flip rate bands and deployed-suboptimal share bands.

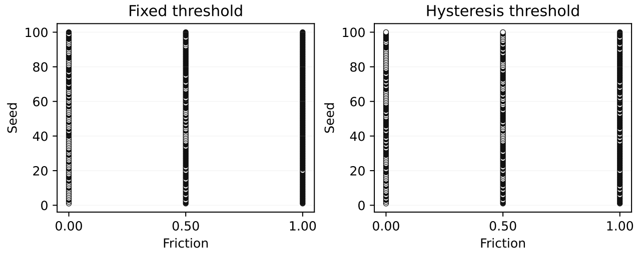

Figure A6: Interface-aware seed-level disagreement strips for the event-micro benchmark. Each panel shows the same seed set under one fixed interface.

E Load-Following as a Second Operational Corroboration Domain

We retain a second operational forecasting domain based on aggregated electricity load-following only as appendix-only secondary corroboration. Forecasts are translated into operational actions under a fixed responsive interface, so this section follows the event-micro robustness package as a diagnostic appendix support block rather than as additional headline evidence. Using UCI ElectricityLoadDiagrams20112014, we build an aggregate multi-client load-following domain by summing disjoint client groups and evaluating at the selected 60-minute operational resolution. Groups 0 and 1 are used only for restricted calibration, while groups 2–9 serve as held-out evaluation units, so this appendix replaces the earlier household proxy with a stronger cross-sectional operational design.

A grouping-only balance-repair rerun substantially reduces group imbalance and Q1 target clipping while preserving the Q2 drift structure. A final restricted cap/margin search, however, finds no jointly feasible configuration that also satisfies the remaining Q1 requirement, so the domain remains appendix-only secondary corroboration rather than additional Q1 promotion evidence.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Linear AR	Small MLP	0.70	0.000	0.000	3/10
0.25	Linear AR	Small MLP	0.30	0.001	0.000	7/10
0.50	Linear AR	Small MLP	0.30	0.001	0.001	7/10
1.00	Linear AR	Small MLP	0.20	0.002	0.002	8/10

Table A11: Load-following Q2 summary retained as appendix-only secondary corroboration under a fixed responsive interface.

This mixed zero row is not a tie-policy artifact: the zero-row winning sets are singletons in all seeds, yet the locked candidate still shows $3/10$ zero-friction mismatches with mean deployed gap 0.0002 and median gap 0.0. The grouping-only balance repair lowers the zero-row mean gap to about 0.0001 but does not remove the mismatch, so the most plausible reading is near-zero seed instability or grouping granularity rather than a clean zero-friction story.

From friction 0.25 onward, the forecast-side winner remains Linear AR while the deployed winner is Small MLP, agreement stays at 0.30, 0.30, and 0.20, and deployed-suboptimal counts rise to $7/10$, $7/10$, and $8/10$. The domain therefore contributes directionally consistent secondary corroboration only; it does not serve as headline evidence or as additional Q1 promotion evidence in the current round.

F Stronger Inventory Q2 Baselines

To reduce baseline-specification ambiguity, we widen the inventory Q2 comparison set with stronger classical and neural families while leaving the environment, split, friction grid, fixed responsive interface, deployed metric, and tie policy unchanged. This later appendix section is included as a stricter-gate transparency layer rather than as earlier or stronger headline evidence than the locked five-family inventory table. All tunable families share the same held-out protocol, validation criterion, and standardized two-stage tuning structure, while the two simple heuristics remain fixed ex ante. Family representatives are selected by validation negative MAE only, and the held-out forecast-side winners in the robustness table are then computed from the same held-out forecast metric used in the locked inventory Q2 pipeline.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Gradient-boosted trees	0.60	0.043	4/10
0.25	Small MLP	Gradient-boosted trees	0.60	0.059	4/10
0.50	Small MLP	Gradient-boosted trees	0.50	0.128	5/10
1.00	Small MLP	Moving average (7)	0.10	0.684	9/10

Table A14: Expanded-family inventory Q2 robustness summary for the broader comparison set.

In this broader comparison, agreement is mixed at 0.00 and 0.25, falls further by 0.50 and 1.00, deployed-suboptimal counts rise from $4/10$ to $9/10$, and moderate/high-friction mismatch strengthens. Because this block uses the stricter package gate, the mixed zero/low-friction rows do not overturn the narrower main claim from the locked five-family table.

G Why Forecast Ordering Need Not Be Preserved

Proposition. A proper forecast-side score need not be order-preserving for deployed utility under a fixed frictional action interface.

Reversal condition. If a forecaster's forecast-side score advantage is smaller than the extra friction-induced utility loss from its induced action path, deployed ordering can reverse.

Proof sketch. Properness concerns truthful prediction scoring. Deployed model selection uses the composed object forecast $\rightarrow$ action proposal $\rightarrow$ interface $\rightarrow$ realized utility. Therefore forecast-side ordering need not be preserved after action mediation.

Family	Input	Stage 1	Stage 2	Seeds	Representative rule
Naive persistence	fixed heuristic	–	–	final 10	fixed ex ante
Moving average (7)	fixed heuristic	–	–	final 10	fixed ex ante
Linear AR	legacy tabular	5	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Small MLP	legacy tabular	18	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Small GRU	sequence (lookback 28)	6	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Regularized linear + lag search	lagged tabular	36	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Gradient-boosted trees	lagged tabular	216	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Large MLP	legacy tabular	108	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
GRU variant	sequence	81	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config

Table A12: Standardized tuning protocol for the stronger-baseline comparison. All tunable families share the same split, validation criterion, and two-stage selection structure; the two simple heuristics remain fixed ex ante.

Family	Representative	Validation mean	Validation median
Naive persistence	fixed ex ante	-2.911	-2.981
Moving average (7)	fixed ex ante	-3.265	-3.251
Linear AR	alpha=0.0001	-2.449	-2.512
Small MLP	lr=0.001, wd=0.0, batch=32	-2.331	-2.390
Small GRU	lr=0.001, batch=32	-2.628	-2.571
Regularized linear + lag search	lag=7, lasso, alpha=0.1	-2.418	-2.410
Gradient-boosted trees	lag=14, trees=100, depth=5, lr=0.1, leaf=10	-2.425	-2.279
Large MLP	w=256, d=3, drop=0.0, lr=0.001, wd=0.0001	-1.954	-2.012
GRU variant	L=28, h=128, layers=2, drop=0.0, lr=0.001	-2.602	-2.599

Table A13: Validation-selected representatives for the broader inventory Q2 family sweep. Family representatives are chosen by validation negative MAE only.